

Experiment No.	05
Experiment Title	Decision Tree and Random Forest Classification
GitHub Repository:	https://github.com/KJayasuriya/ml_laboratory/
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1 Objective

To classify breast cancer cases as Malignant or Benign using Decision Tree and Random Forest classification algorithms. The experiment also aims to compare the performance of the two models, tune their hyperparameters using Grid Search with 5-Fold Cross-Validation, and evaluate their classification performance using standard evaluation metrics.

2 Problem Statement

Develop machine learning classification models to predict whether a breast tumor is Malignant (M) or Benign (B) using numerical features extracted from digitized images of breast mass samples. A Decision Tree classifier and a Random Forest classifier are trained on the dataset and their performance is compared before and after hyperparameter tuning.

3 Dataset Description

The experiment uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The dataset contains numerical measurements computed from digitized images of breast mass samples. The target variable, **diagnosis**, contains two classes: Malignant (M) and Benign (B).

The dataset used in the notebook is loaded from `../datasets/breastCancer/wdbc.data`. The notebook assigns 32 columns: one identifier column, one diagnosis column, and 30 numerical measurement features grouped into mean, standard error, and worst-case measurements.

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository / WDBC dataset
Number of Samples	569
Number of Columns	32
Number of Predictive Features	30 numerical features
Number of Classes	2 (Malignant - M, Benign - B)
Missing Values	0
Duplicate Rows	0
Train-Test Split	Train - 455, Test - 114

The class distribution obtained from the notebook is:

- Benign (B): 357 samples
- Malignant (M): 212 samples

Thus, approximately 62.74% of the samples are Benign and 37.26% are Malignant. The dataset has a moderate class imbalance.

4 Exploratory Data Analysis

Dataset Structure and Data Quality

The dataset contains 569 samples and 32 columns. The notebook confirms that all measurement columns are numerical and that the diagnosis column is a string categorical variable. No missing values or duplicate rows were found.

The numerical features represent measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Each measurement is available in three forms: mean, standard error, and worst value.

Class Distribution

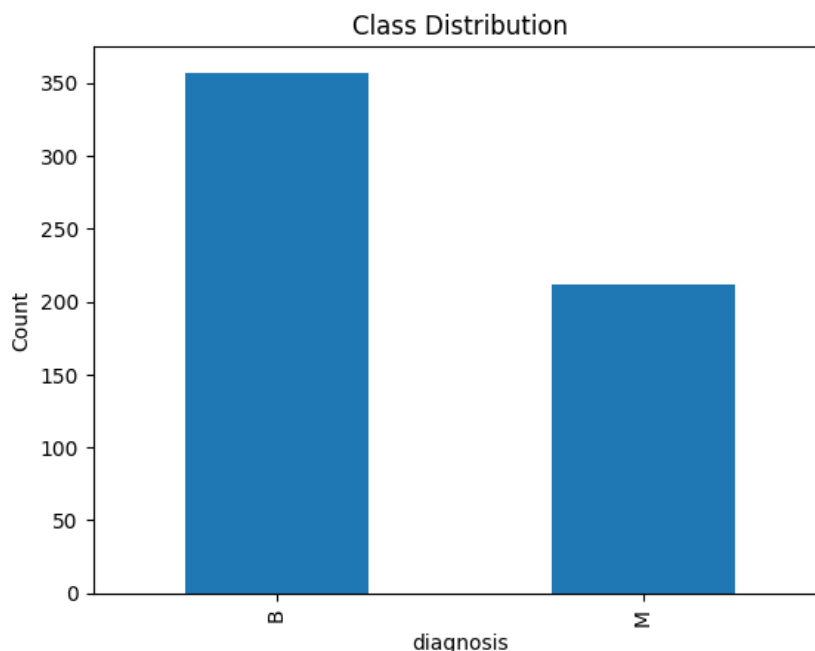


Figure 1: Class Distribution of Diagnosis

Inference:

The class distribution shows that the dataset contains 357 Benign cases and 212 Malignant cases. Therefore, the Benign class represents approximately 62.74% of the

dataset, while the Malignant class represents approximately 37.26%. The classes are not perfectly balanced, but the imbalance is not extreme. Hence, both classes have sufficient samples for training and evaluation.

Correlation Matrix

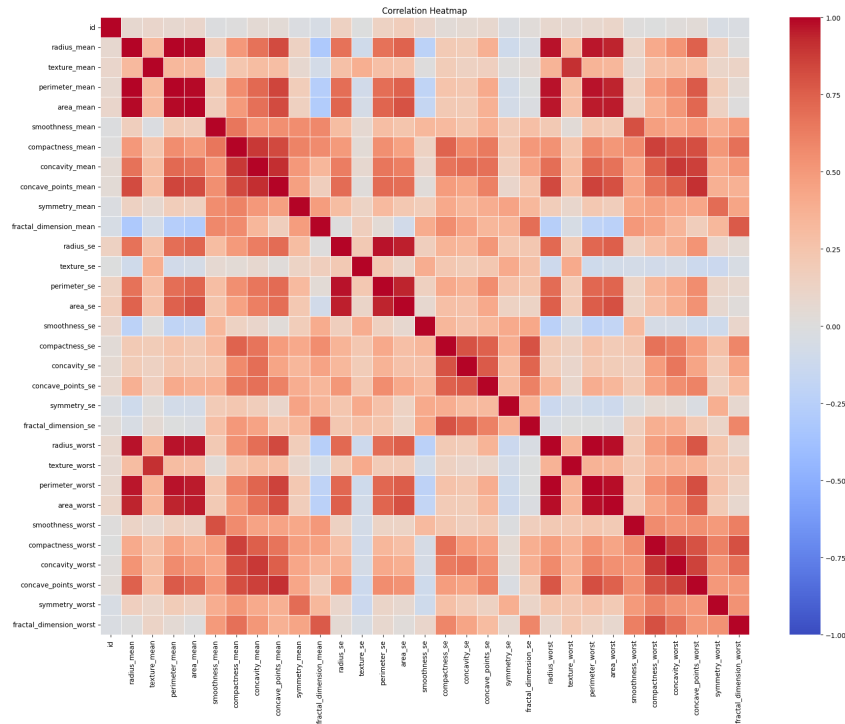


Figure 2: Correlation Matrix of Numerical Features

Inference:

The correlation matrix shows the relationships among the numerical features in the breast cancer dataset. Several measurements, particularly related features such as radius, perimeter, and area, exhibit strong positive correlations. This indicates that some features contain overlapping information. Such relationships are useful for tree-based classifiers because Decision Trees and Random Forests do not require the features to be independent.

5 Data Preprocessing

- **Missing Value Handling:** The dataset contains no missing values, so no imputation or removal of samples was required.
- **Duplicate Handling:** No duplicate rows were detected in the dataset.
- **Target Variable:** The target variable is `diagnosis`, containing the classes Malignant (M) and Benign (B). Scikit-learn classifiers can use these string class labels directly.
- **Feature Scaling:** Feature scaling was not performed. Decision Tree and Random Forest classifiers are tree-based methods and therefore do not require normalization or standardization of numerical features.

- **Train-Test Split:** The dataset was divided into training and testing sets using an 80:20 split with `random_state=42`. This resulted in 455 training samples and 114 testing samples.
- **Feature Engineering:** No additional feature engineering was performed because the dataset already contains extracted numerical features.
- **Identifier Column:** The implementation used in the notebook removes only the `diagnosis` column before training. Therefore, the `id` column is also included in the feature matrix. This is a limitation of the current implementation because the identifier is not a meaningful biological feature.

6 Machine Learning Algorithm

6.1 Decision Tree

A Decision Tree is a supervised learning algorithm that recursively divides the dataset into smaller subsets using decision rules based on feature values. At each internal node, the algorithm selects a feature and threshold that produces the best separation between the target classes.

For classification, the quality of a split can be measured using impurity criteria such as Gini impurity or entropy.

The Gini impurity is given by

$$G = 1 - \sum_{i=1}^K p_i^2$$

where p_i represents the proportion of samples belonging to class i .

Entropy is defined as

$$H = - \sum_{i=1}^K p_i \log_2(p_i)$$

A split is selected such that the impurity of the resulting child nodes is reduced.

The Decision Tree continues splitting until stopping conditions such as maximum depth or minimum number of samples are reached.

Advantages

- Easy to understand and interpret.
- Does not require feature scaling.
- Can model nonlinear relationships.
- Can handle numerical features directly.

Limitations

- A deep tree can easily overfit the training data.
- Small changes in the training data can produce a different tree.
- A single tree may have lower generalization performance than an ensemble method.

6.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees. Each tree is trained using a randomly selected subset of the training data and a random subset of features.

For classification, every tree produces a class prediction and the Random Forest selects the class receiving the majority of votes.

If T trees predict classes $h_1(x), h_2(x), \dots, h_T(x)$, the final prediction can be represented as

$$\hat{y} = \text{mode} \{h_1(x), h_2(x), \dots, h_T(x)\}.$$

The use of multiple trees reduces the variance of an individual Decision Tree and generally improves generalization.

Advantages

- Usually provides better generalization than a single Decision Tree.
- Reduces the risk of overfitting through ensemble learning.
- Can model complex nonlinear relationships.
- Does not require feature scaling.
- Provides feature importance information.

Limitations

- More computationally expensive than a single Decision Tree.
- Less interpretable than an individual Decision Tree.
- The model requires more memory because multiple trees are stored.

7 Experimental Setup

Python Version	3.13.13
Scikit-Learn Version	1.7.5
IDE	Jupyter Notebook
Operating System	Fedora 42
Hardware	Intel Core i5 (11th gen), 8GB RAM

8 Hyperparameter Selection

Grid Search with 5-Fold Cross-Validation was used to select suitable hyperparameters for both classifiers.

Decision Tree

The following hyperparameters were searched:

- `criterion`: gini, entropy
- `max_depth`: 3, 5, 10, None
- `min_samples_split`: 2, 5, 10
- `min_samples_leaf`: 1, 2, 4

The best Decision Tree configuration was:

Parameter	Value
Criterion	entropy
Max Depth	5
Minimum Samples Split	2
Minimum Samples Leaf	1

The best 5-Fold Cross-Validation accuracy was 0.9473.

Random Forest

The following hyperparameters were searched:

- `n_estimators`: 50, 100, 200
- `max_depth`: 5, 10, None
- `min_samples_split`: 2, 5
- `min_samples_leaf`: 1, 2

The best Random Forest configuration was:

Parameter	Value
Number of Estimators	200
Max Depth	10
Minimum Samples Split	2
Minimum Samples Leaf	2

The best 5-Fold Cross-Validation accuracy was 0.9604.

9 Results

The final models were evaluated on the independent test set containing 114 samples.

Metric	Decision Tree	Random Forest
Best CV Accuracy	0.9473	0.9604
Test Accuracy	0.9561	0.9649
Weighted Precision	0.9569	0.9652
Weighted Recall	0.9561	0.9649
Weighted F1-score	0.9558	0.9647
ROC-AUC	Not calculated	Not calculated

The baseline Decision Tree obtained a test accuracy of 0.947. After hyperparameter tuning, the Decision Tree achieved a test accuracy of 0.956.

The tuned Random Forest achieved a test accuracy of 0.965, with weighted precision of 0.9652, weighted recall of 0.9649, and weighted F1-score of 0.9647.

The Random Forest therefore performed better than the tuned Decision Tree on the test dataset.

10 Result Visualization

Class Distribution

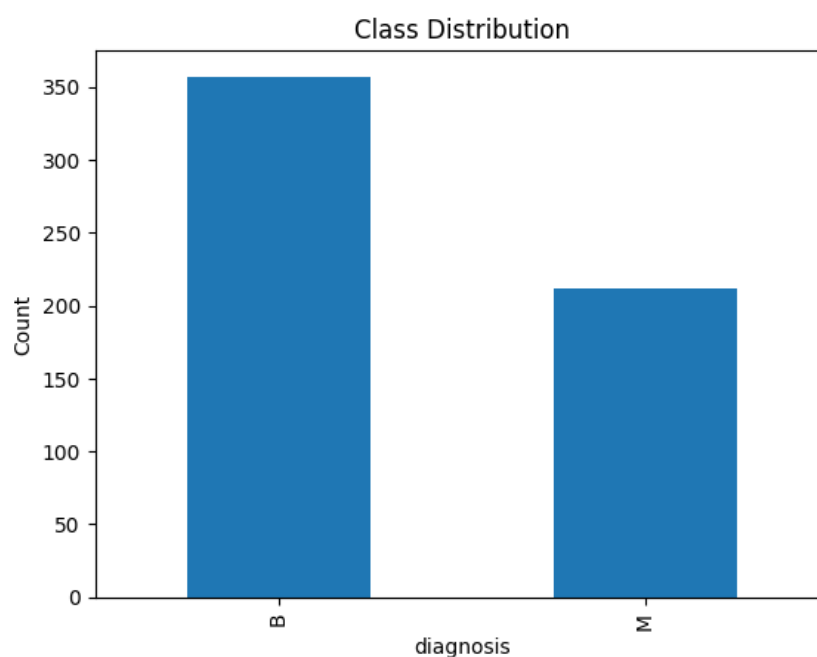


Figure 3: Class Distribution of Diagnosis

Inference:

The dataset contains more Benign than Malignant samples. There are 357 Benign samples and 212 Malignant samples. The moderate class imbalance should be considered when interpreting classification metrics. However, both classes contain a substantial number of observations.

Correlation Matrix

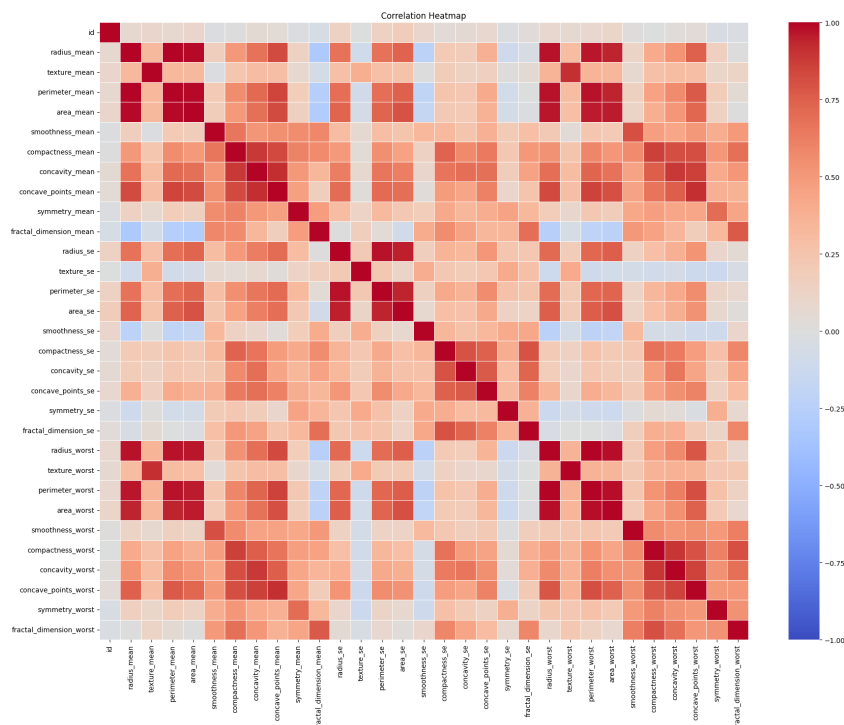


Figure 4: Correlation Matrix of Numerical Features

Inference:

The correlation matrix demonstrates strong relationships among several measurements. Features such as radius, perimeter, and area are strongly related because they describe related physical properties of the cell nucleus. The presence of correlated features does not prevent tree-based models from being trained.

Model Performance Comparison

The following values were obtained directly from the executed notebook:

Model	CV Accuracy	Test Accuracy	Precision	Recall	F1-score
Decision Tree	0.9473	0.9561	0.9569	0.9561	0.9558
Random Forest	0.9604	0.9649	0.9652	0.9649	0.9647

Inference:

The Random Forest achieved better cross-validation and test accuracy than the Decision Tree. Its weighted F1-score was also higher, indicating better overall classification performance. The improvement is consistent with the ability of Random Forest to combine multiple trees and reduce the variance associated with a single Decision Tree.

Note: The current notebook does not generate confusion matrices, ROC curves, Precision-Recall curves, learning curves, or a feature-importance plot. Therefore, these figures are not included as experimental results.

11 Discussion

The experiment demonstrates the application of Decision Tree and Random Forest classifiers to breast cancer classification. The baseline Decision Tree achieved a test accuracy of 0.947. Hyperparameter tuning improved its test accuracy to 0.956, showing that controlling tree complexity can improve generalization.

The tuned Decision Tree selected entropy as the splitting criterion and a maximum depth of 5. Limiting the depth prevents the tree from becoming unnecessarily complex and helps reduce overfitting.

The Random Forest achieved a higher 5-Fold Cross-Validation accuracy of 0.9604 and a test accuracy of 0.9649. The model used 200 trees, a maximum depth of 10, and a minimum of two samples per leaf. The ensemble of multiple trees provided better generalization than the single Decision Tree.

The results demonstrate the bias–variance advantage of ensemble methods. While an individual Decision Tree is simple and interpretable, it can have high variance. Random Forest reduces this variance by averaging predictions from many independently constructed trees.

A limitation of the current implementation is that the identifier column `id` is included in the feature matrix. Since this identifier does not represent a meaningful biological measurement, it should ideally be removed before model training. Another limitation is that ROC-AUC and class-specific confusion-matrix analysis were not calculated in the current notebook.

Future improvements could include removing the identifier column, using stratified splitting, reporting confusion matrices, calculating ROC-AUC and Precision-Recall curves, and analyzing Random Forest feature importance.

12 Conclusion

The experiment successfully implemented Decision Tree and Random Forest classifiers for binary breast cancer classification. Grid Search with 5-Fold Cross-Validation was used to identify suitable hyperparameters for both models.

The tuned Decision Tree achieved a test accuracy of 0.9561, while the tuned Random Forest achieved a higher test accuracy of 0.9649. The Random Forest also obtained higher weighted precision, recall, and F1-score.

Therefore, among the two models evaluated in this experiment, Random Forest provided the best overall classification performance. The experiment also demonstrated the importance of hyperparameter tuning and ensemble learning for improving the generalization performance of classification models.

13 References

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5. Scikit-learn Documentation, “Decision Trees,” Scikit-learn.
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